1 Question (16 points)

Are the following languages regular? Prove your answer.

a. Let A be regular, then the language $A^R = \{w^R | w \in A\}$, where w^R is the string written w in reverse order.

b. $L = \{w | w = w^R\}$, where w^R is the string written w in reverse order.

2 Question (16 points)

Are the following languages context-free? Prove your answer.

- A $= \{a^n b^m c^p d^p e^m f^n | n, m, p \geq 0\}$
- B $= \{a^n b^m c^m d^n e^p f^p | n, m, p \geq 0\}$

3 Question (18 points)

Design a TM that recognizes the language $L = \{a^n b^m c^m d^n | m, n \geq 1\}$. Do not describe the steps of TM. Draw a state diagram, give states, transitions, etc.

4 Question (16 points)

Describe a TM that recognizes the language $L = \{a^n b^m c^{n-m} | n, m \geq 1, n \geq m\}$. Do not design the TM. Describe the steps it needs to take.

5 Question (16 points)

Describe a multitape Turing machine with $k+1$ tapes that recognizes the language $L = \{x_1^1 x_2^2 \ldots x_k^k | k \geq 1\}$, $\Sigma = \{x_1, x_2, \ldots, x_k\}$. You need to use all tapes in your solution. Do not design the TM. Describe the steps it needs to take.

6 Question (18 points)

Is the complement of a decidable language decidable? Prove it.